

Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Machine cognition — the capacity of computers to replicate mental faculties such as pattern recognition, logical inference, and adaptive learning — is the core idea behind AI. Systems built on these principles handle challenges like decoding spoken words, identifying objects in photos, and selecting optimal actions.

Body:

Several technical disciplines fall under the AI umbrella. Statistical modelling from large datasets (machine learning) is foundational. Stacked perceptron architectures (deep networks) excel at high-dimensional perception tasks. Computational linguistics (NLP) bridges human and machine communication. Reward-guided trial and error (reinforcement learning) trains agents without labelled data. These methods power radiology assistants, transaction monitoring systems, and highway autopilots.

Conclusion:

Synthetic cognition is the defining technological shift of our era. The productivity gains are real, but so are the risks: skewed training sets encode social inequity; ubiquitous surveillance erodes privacy; labour displacement strains social contracts. Ensuring algorithmic accountability and interpretability is the prerequisite for harnessing these systems safely.